

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (5 points) Evaluate the following indefinite integrals:

(a) $\int x\sqrt{x^2+2} dx$

Let $u = x^2 + 2$
 $du = 2x dx$

$\frac{du}{2} = x dx$

Thus

$$\begin{aligned}\int x\sqrt{x^2+2} dx &\stackrel{u=x^2+2}{=} \int \frac{\sqrt{u}}{2} du \\ &= \frac{1}{2} \cdot \frac{u^{3/2}}{3/2} + C \\ &= \frac{1}{3} u^{3/2} + C \\ &= \frac{1}{3} (x^2+2)^{3/2} + C\end{aligned}$$

(b) $\int \cos \theta \sin^4 \theta d\theta$

Let $u = \sin \theta$
 $du = \cos \theta d\theta$

Thus

$$\begin{aligned}\int \cos \theta \sin^4 \theta d\theta &\stackrel{u=\sin \theta}{=} \int u^4 du \\ &= \frac{u^5}{5} + C \\ &= \frac{1}{5} \sin^5 \theta + C\end{aligned}$$

Problem 2. (5 points) Evaluate the following definite integrals:

(a) $\int_1^{e^2} \frac{\ln x}{x} dx$

Let $u = \ln x$, then $du = \frac{1}{x} dx$

$x=1 \rightarrow u = \ln 1 = 0$

$x=e^2 \rightarrow u = \ln e^2 = 2$

$$\begin{aligned}\Rightarrow \int_1^{e^2} \frac{\ln x}{x} dx &\stackrel{u=\ln x}{=} \int_0^2 u du \\ &= \left[\frac{u^2}{2} \right]_{u=0}^{u=2} \\ &= \frac{2^2}{2} - \frac{0^2}{2} \\ &= 2\end{aligned}$$

(b) $\int_0^1 x e^{-x^2} dx$

Let $u = -x^2$, then $du = -2x dx$

$-\frac{du}{2} = x dx$

$x=0 \rightarrow u = -0^2 = 0$

$x=1 \rightarrow u = -1^2 = -1$

$$\begin{aligned}\Rightarrow \int_0^1 x e^{-x^2} dx &\stackrel{u=-x^2}{=} \int_0^{-1} -\frac{e^u}{2} du \\ &= \left[-\frac{e^u}{2} \right]_{u=0}^{u=-1} \\ &= \left(-\frac{e^{-1}}{2} \right) - \left(-\frac{e^0}{2} \right) \\ &= -\frac{1}{2e} + \frac{1}{2}\end{aligned}$$